GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Tableau Data Visualization Project Documentation

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Primary tool	Tableau

Before submission, replace the bracketed placeholders and add verified findings and project links.

1. Executive Summary

This project explores historical Olympic performance through interactive Tableau visualizations. It examines medal outcomes by sport, country and region; participation changes over time; and host-nation performance. The project is designed around three perspectives: national sports strategy, sports investment analysis, and historical storytelling.

2. Problem Statement

Olympic datasets contain records across many years, countries, sports and events. Reviewing raw tables can make it difficult to recognize long-term trends or compare performance consistently. This project uses data visualization to organize the information and communicate patterns for exploration and reporting.

3. Project Objectives

• Analyze medal performance across countries and sport categories.

• Compare medal outcomes with athlete participation to explore medal efficiency.

• Examine regional differences in medal distribution and sport performance.

• Track athlete and nation participation over Olympic years.

• Explore host-nation medal performance in the relevant host editions.

• Present findings through dashboards and a Tableau story.

4. Analysis Scenarios

Scenario 1 — National Olympic Committee strategist: Compare sports and countries to identify categories with comparatively strong medal performance. Average medals per athlete can provide an efficiency-oriented view and help identify areas for further investigation.

Scenario 2 — Sports investment analyst: Compare medal outcomes across sports and regions and examine medal efficiency. If the dataset does not contain financial expenditure, this is a performance-efficiency analysis, not a direct monetary return-on-investment calculation.

Scenario 3 — Broadcast journalist / historical storyteller: Use participation trends, host-nation comparisons and regional or sport-level medal distributions to explain how Olympic participation and competitive outcomes have changed over time.

5. Dataset and Data Preparation

Dataset: [Enter exact dataset name, source and URL]. The analysis uses Olympic-related records such as year, season, country/NOC, sport, athlete participation and medal information, depending on the fields available in the chosen dataset.

Before final submission, verify field names and data types, standardize country and sport labels, inspect null values, and understand the row-level grain to avoid accidental double-counting. For medals per athlete, define the medal numerator and distinct-athlete denominator at the same country/sport/time level. Explain how team events and multiple medals are counted.

6. Visualizations

No.	Visualization	Purpose
1	Average Medals per Athlete by Sport	Compares medal output relative to athlete participation across sports.
2	Country / Sport Performance Comparison	Compares country performance across sport categories.

No.	Visualization	Purpose
3	Medal Performance by Country	Ranks or compares countries by medal outcomes.
4	Medal Efficiency by Region	Compares regional medal efficiency using the project's defined ratio.
5	Regional Medal Distribution	Shows how medal outcomes are distributed among regions.

6	Regional Medal Distribution by Sport	Breaks down regional medal distribution across sport categories.
7	Host Nation Medals	Highlights host-nation performance in the relevant host editions.
8	Participation Trends	Tracks participation measures across Olympic years.
9	Olympic Athlete and Nation Participation Trends	Compares athlete and participating-nation trends over time.

Adjust these names and descriptions if your final workbook uses different titles or chart designs.

7. Dashboards and Tableau Story

Dashboard 1 — Olympic performance overview: Bring together country and sport performance, medal comparisons and participation context. Keep filters consistent and add a short caption explaining the dashboard's purpose.

Dashboard 2 — Sports investment vs. medal performance: The discussed layout includes a sport-performance scatterplot and regional medal-efficiency / distribution views. Clearly explain the measures, mark size and color. If a pie chart has many small slices, consider grouping minor categories or using a sorted bar chart.

Tableau Story: Arrange story points in a logical sequence: overall participation and medal context, country and sport comparisons, host-nation performance, then regional and efficiency insights.

8. Key Findings

Complete this section using values verified in your final Tableau workbook:

• Sports with highest / lowest medals per athlete: [Insert sports and values].

• Countries or regions with notable medal performance: [Insert verified examples].

• Participation trend across Olympic years: [Describe observed trend].

• Host-nation performance pattern: [Insert verified comparison].

• Important outlier or limitation: [Explain].

Do not infer causation from these charts. Results can be affected by event counts, qualification systems, historical participation, dataset coverage and medal-counting rules.

9. Tools and Technologies

• Tableau Desktop / Tableau Public — data exploration, calculated fields, worksheets, dashboards and story.

• Olympic dataset — source data for participation, country, sport, year and medal analysis. • GitHub — project file storage and documentation.

• Other tools used: [List if applicable].

10. Limitations

Conclusions depend on the coverage and definitions in the selected dataset. Medal totals may be affected by team events, shared medals, duplicate athlete records and country/NOC changes. Medal efficiency is a descriptive ratio, not a measure of coaching quality or financial return. If financial investment data is absent, the

project cannot establish monetary return on investment. Host-nation comparisons should use the host nation's actual host edition rather than all years combined.

11. Conclusion

The project demonstrates how Tableau can turn Olympic records into interactive views of participation, medal outcomes and regional or sport-level patterns. The dashboards support strategic exploration and historical storytelling. Results are most useful when metric definitions are clear and comparisons are interpreted in light of dataset limitations.